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The Riemann Hypothesis manifested in dynamical quantum phase transitions

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The Riemann Hypothesis (RH), one of the most profound unsolved problems in mathematics, concerns the nontrivial zeros of the Riemann zeta function. Linking these zeros to physical phenomena offers new perspectives on their origin and verification. Here we establish a direct correspondence between these zeros and dynamical quantum phase transitions in two complementary engineered quantum many-body systems, characterized by the average accumulated phase factor and the Loschmidt amplitude, respectively. This precise correspondence recasts the RH as the occurrence of phase transitions at a unique temperature and identifies it as a previously unknown transition mechanism. We demonstrate this correspondence in a proof-of-principle experiment on a quantum processor. Moreover, we propose a quantum computational framework that implements both systems with polynomial resources, suggesting quantum advantage in probing the hypothesis. Our work bridges nonequilibrium quantum dynamics and number theory, positioning quantum computing as a powerful platform for exploring mathematical conjectures, phase transitions and beyond.

The Riemann zeta function is central to analytic number theory and has profound implications in mathematical physics. It admits a Dirichlet series representation valid for $\mathrm{Re}(s) > 0, \mathrm{Re}(s) \neq 1$[1]:

$$\zeta(s) = \frac{1}{1-2^{1-s}} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^s}. \tag{1}$$

The nontrivial zeros of the zeta function, conjectured by the celebrated Riemann Hypothesis (RH) to all lie on the critical line $\mathrm{Re}(s) = \frac{1}{2}$, remain one of the most profound unsolved problems in mathematics. The RH underpins more than a thousand theorems in mathematics and forms the foundation of the modern RSA cryptographic system; for example, the distribution of the nontrivial zeros encodes the fine structure of prime numbers[2]. Despite extensive numerical evidence[3,4], the RH remains unproven. Any new perspective on the zeta function is therefore of great significance, regardless of whether it ultimately leads to a proof or refutation of the RH.

Beyond its mathematical significance, the Riemann zeta function has long inspired deep physical analogies[5], sparking the search for a possible physical origin of the RH. The most influential idea in this direction is the Hilbert-Pólya conjecture[6,7], which suggests that the nontrivial zeros of the zeta function might correspond to the eigenvalues of an as-yet-unknown quantum Hamiltonian, motivating searches for such an operator[8,9] and for signatures in diverse physical contexts[5,10], including relativistic amplitudes[11], chaotic quantum scattering[12,13], quantum field theory[14], and random matrix theory[15]. Significant progress[16–18] has linked quantum systems to the Riemann zeta function through tailored state preparation, engineered spectra, and observable correspondences with its zeros. The genuine physical realization, deeper connections and possible origins of the RH remain unknown.

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Here, we propose a physical origin for the RH by demonstrating that the nontrivial zeros of the Riemann zeta function correspond directly to dynamical quantum phase transitions (DQPTs) in two complementary quantum systems within a framework[19–26], as shown in Fig. 1. Both systems are initialized in thermal equilibrium and subsequently driven out of equilibrium by quenches with specially designed interaction Hamiltonians that imprint properties of the zeta function onto measurable observables. The average accumulated phase factor in the first system and the Loschmidt amplitude in the second system both vanish at evolution times corresponding to the nontrivial zeros, revealing DQPTs. By calculating the free energy and entropy, we further investigate the characteristics of the DQPTs.

Furthermore, we introduce a quantum computational framework that efficiently simulates these dynamics on digital quantum processors, offering a clear quantum advantage for probing both DQPTs and the RH. This approach not only enables the experimental observation of DQPTs tied to the zeros but also allows for the numerical exclusion of such critical behavior at non-zero points, thereby providing a physical pathway to investigate the RH.

As a proof-of-principle demonstration, we implement the first system experimentally on a five-qubit nuclear spin quantum processor. The measured coherence dynamics reproduce the oscillatory behavior of the zeta function and exhibit clear signatures of DQPTs aligned with its nontrivial zeros. For the second system, suitable for exploring high-index zeros, large-scale numerical simulations confirm the correspondence up to the 10^{12}-th zero. Moreover, our gate-based framework constructs both models with polynomial resources and achieves at least a quadratic speedup over classical verification of RH zeros. These results establish a concrete bridge between number theory, quantum physics, and quantum technology, opening scalable pathways toward one of mathematics' most enduring mysteries and hinting at a deeper quantum structure underlying arithmetic.

Results

As shown in Fig. 1c, we study two complementary quantum dynamical processes within a framework, each initialized in a thermal state. In both cases, a single probe qubit is coupled to the system, and the combined system evolves under an interaction Hamiltonian for a duration t. The average accumulated phase factor $\mathcal{L}(\beta, t)$ in the first setup directly encodes the behavior of $\zeta(s)$ in the thermodynamic limit, while the Loschmidt amplitude $\mathcal{G}(\beta, t)$ in the second captures the same feature in the joint thermodynamic and long-time limits.

Remarkably, at $\beta = \frac{1}{2}$, both systems exhibit DQPTs, characterized by recurrent vanishing-and-revival oscillations in the observables, which correspond one-to-one to the nontrivial zeros. If the RH holds, such recurrent revivals occur exclusively on the critical line $\beta = \frac{1}{2}$ in the first system. This correspondence indicates that RH can be reformulated as the statement that DQPTs occur only at the inverse

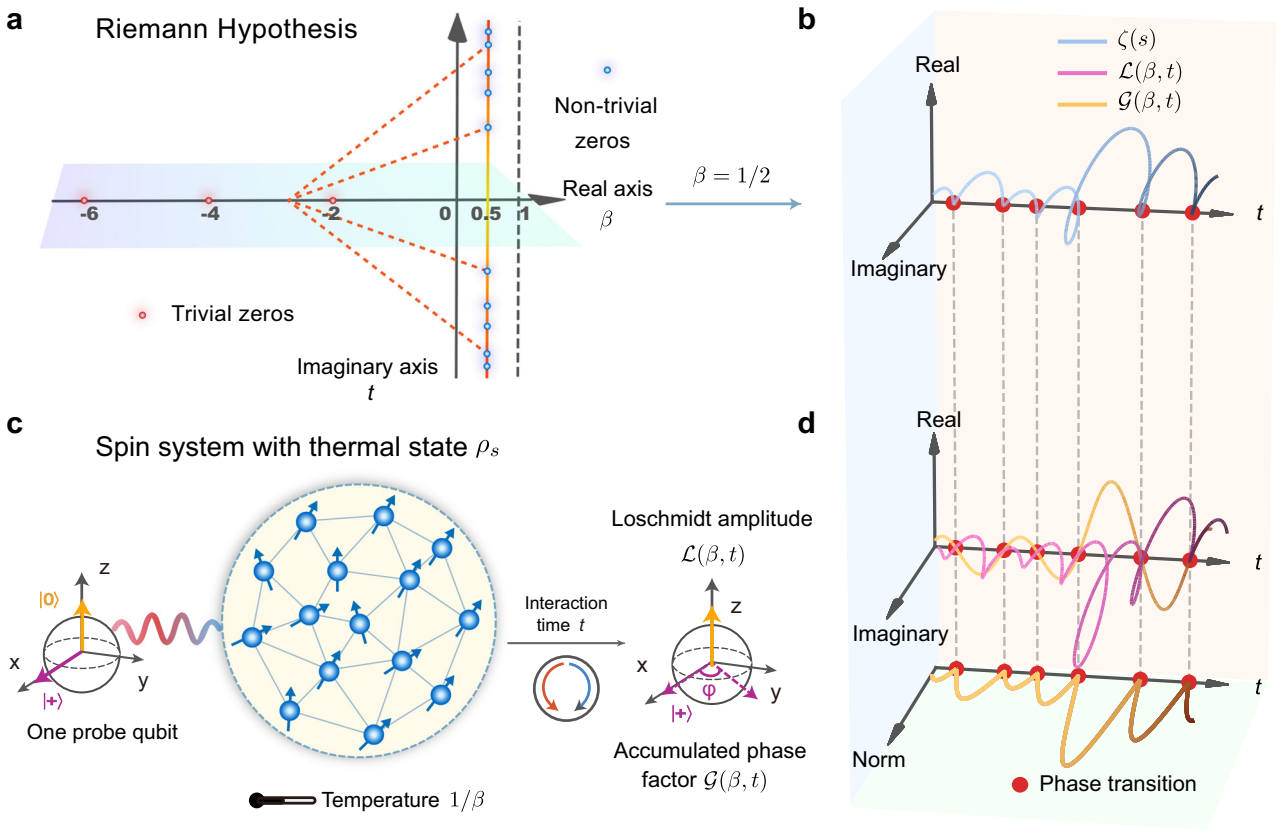


Fig. 1 | Correspondence between the Riemann zeta function and engineered quantum many-body systems. a The RH in the complex plane (β, t). It states that all nontrivial zeros of the zeta function lie on the critical line with real part $\beta = 1/2$. The distribution of the zeros of the zeta function is symmetric with respect to reflection across the real axis. **b** The zeta function $\zeta(1/2 + it)$ along the critical line. Each zero crossing, marked by a red dot, corresponds to a nontrivial zero of the zeta function. **c** Construction of two complementary quantum systems within a framework, each consisting of a single probe spin coupled to a many-body system initialized in a thermal state at inverse temperature β. The composite system subsequently evolves under the system interactions with time-reversal symmetry. The blue and red arrows represent the forward and backward time evolutions, respectively. **d** The dynamics of the accumulated phase factor $\mathcal{L}(\beta, t)$ and the Loschmidt amplitude $\mathcal{G}(\beta, t)$ reproduce key features of the zeta function, exhibiting vanishing-and-revival oscillations characteristic of DQPTs. Red dots highlight the phase-transition points, in one-to-one correspondence with the nontrivial zeros of the Riemann zeta function. The norms of both observables correspond to the norm of the zeta function.

temperature $\beta=\frac{1}{2}$ in a specific quantum many-body system. Conversely, the RH reveals a previously unknown class of zeta-governed mechanisms for DQPTs. Moreover, the quantum systems exhibit time-reversal symmetry, which corresponds to the reflection symmetry of the zeta function. Together, these results establish a direct correspondence between the analytic features of the zeta function (Fig. 1b) and accessible quantum dynamics (Fig. 1d). We describe the construction of these quantum dynamics in the following sections.

Probe spin coherence signaling RH

We consider a quantum system with physical Hamiltonian

$$\mathcal{H}_0^{\mathrm{ph}}=\sum_{n=1}^{N}E_n|n\rangle\langle n|,\qquad E_n=\varepsilon_0\ln n, \tag{2}$$

where ε_0 sets the energy scale[16]. The system is initialized in the thermal state $\rho_s=e^{-\mathcal{H}_0^{\mathrm{ph}}/(k_BT)}/\mathcal{Z}(T,\mathcal{H}_0^{\mathrm{ph}})$, with partition function

$$\mathcal{Z}(T,\mathcal{H}_0^{\mathrm{ph}})=\sum_{n=1}^{N}e^{-E_n/(k_BT)}=\sum_{n=1}^{N}n^{-\varepsilon_0/(k_BT)}. \tag{3}$$

We then introduce the dimensionless inverse temperature, Hamiltonian, and time,

$$\beta\equiv\frac{\varepsilon_0}{k_BT},\qquad \mathcal{H}_0\equiv\frac{\mathcal{H}_0^{\mathrm{ph}}}{\varepsilon_0},\qquad t\equiv\frac{\varepsilon_0 t_{\mathrm{ph}}}{\hbar}. \tag{4}$$

With these definitions, the thermal weight and dynamical phase of the n-th level become

$$e^{-E_n/(k_BT)}=n^{-\beta},\qquad e^{-iE_nt_{\mathrm{ph}}/\hbar}=n^{-it}, \tag{5}$$

and the partition function reduces to

$$\mathcal{Z}(\beta,\mathcal{H}_0)=\sum_{n=1}^{N}n^{-\beta}. \tag{6}$$

The complex parameter $s=\beta+it$ is therefore a natural dimensionless variable. In the following, energies and times are measured in units of ε_0 and $\hbar/\varepsilon_0$, respectively.

Constructing a logarithmic energy spectrum is central to realizing the proposed quantum system. Such spectra have appeared previously in a variety of number-theoretic and quantum-dynamical settings, including Bose–gas analogies for factorization[27], wave-packet constructions related to the zeta function[16], factorization-oriented schemes based on interacting bosons or separable potentials[28,29], and more recent holographic approaches[30]. In most of these works, the logarithmic spectrum is generated through analog constructions, typically by designing specific potentials or effective physical landscapes. Equivalently, for finite N, the logarithmic spectrum admits an exact representation in terms of a many-body spin Hamiltonian. For $N=2^q$ levels, encoding the energy eigenstates into the computational basis of q spins yields

$$\mathcal{H}_0=\sum_{l=0}^{q}\sum_{1\le m_1<\cdots<m_l\le q}J_{m_1\cdots m_l}\,\sigma_{m_1}^{z}\cdots\sigma_{m_l}^{z}, \tag{7}$$

with $l=0$ corresponding to the identity Pauli operator. Writing $n-1=\sum_{m=1}^{q}2^{m-1}b_m^{(n)}$, with $b_m^{(n)}\in\{0,1\}$ and defining $z_m^{(n)}\equiv(-1)^{b_m^{(n)}}$, the coefficients are uniquely determined by

$$J_{m_1\cdots m_l}=\frac{1}{N}\sum_{n=1}^{N}(\ln n)\,z_{m_1}^{(n)}\cdots z_{m_l}^{(n)},\quad l=0,1,\ldots,q. \tag{8}$$

The logarithmic spectrum is therefore not tied to a unique microscopic realization, but admits an exact representation in a spin system. In our experiments, we prepare the corresponding thermal state directly by engineering the level populations of the density matrix on a nuclear spin quantum processor. More generally, we present a universal quantum algorithm that prepares states and Hamiltonians with a logarithmic spectrum to arbitrary precision using only polynomial resources (see Section II D).

We define the average accumulated phase factor as

$$\mathcal{L}(\beta,t)=\mathrm{Tr}\left[\rho_s e^{-i\mathcal{H}_0t}\sigma_z^1\right], \tag{9}$$

where $e^{-i\mathcal{H}_0t}\sigma_z^1$ imprints a phase factor $(-1)^{n+1}n^{-it}$ on the n-th population of ρ_s. This yields

$$\mathcal{L}(\beta,t)=\frac{1}{\mathcal{Z}(\beta,\mathcal{H}_0)}\sum_{n=1}^{N}(-1)^{n+1}\,n^{-\beta-it}\xrightarrow{N\to\infty}\frac{(1-2^{1-(\beta+it)})\,\zeta(\beta+it)}{\mathcal{Z}(\beta,\mathcal{H}_0)} \tag{10}$$

in the thermodynamic limit. Hence, $\mathcal{L}(\beta,t)$ vanishes precisely when $\zeta(\beta+it)$ has a nontrivial zero. These zeros signal DQPTs in mixed states marked by non-analyticities in time [22]. To connect these dynamical transitions to their equilibrium analogues, we invoke the Choi–Jamiołkowski isomorphism. The mixed state ρ_s is mapped to the purification

$$|\rho_s\rangle=\mathcal{Z}(\beta,\mathcal{H}_0)^{-1/2}\sum_{n=1}^{N}n^{-\beta/2}|n\rangle|n\rangle_a \tag{11}$$

in an enlarged Hilbert space $\mathcal{H}\otimes\mathcal{H}_a$, where the ancilla space $\mathcal{H}_a$ is isomorphic to the physical one. Under evolution generated by $\mathcal{H}_0$ on the physical subsystem, the purified state evolves as

$$|\rho(t)\rangle=(e^{-i\mathcal{H}_0t}\otimes I_a)|\rho_s\rangle, \tag{12}$$

and the accumulated phase factor can be written as

$$\mathcal{L}(\beta,t)=\langle\rho_s|e^{-i\mathcal{H}_0t}\sigma_1^z|\rho_s\rangle. \tag{13}$$

In equilibrium boundary statistical mechanics, where $\mathcal{Z}_B(\beta,R,\mathcal{H})=\langle\psi_a|e^{-R\mathcal{H}}|\psi_b\rangle$[19,31], choosing $|\psi_a\rangle=|\rho_s\rangle$, $|\psi_b\rangle=\sigma_1^z|\rho_s\rangle$, and $R=it$ gives

$$\mathcal{Z}_B(\beta,it,\mathcal{H}_0\otimes I_a)=\mathcal{L}(\beta,t), \tag{14}$$

which identifies $\mathcal{L}(\beta,t)$ as the boundary partition function between the boundary states $|\rho_s\rangle$ and $\sigma_1^z|\rho_s\rangle$, analytically continued to complex time.

In the thermodynamic limit, the associated free-energy density is defined as[20,32,33]

$$\begin{aligned}\mathcal{F}_1(\beta,t)&=-\lim_{N\to\infty}\frac{1}{\log N}\ln|\mathcal{Z}_B(\beta,it,\mathcal{H}_0\otimes\mathbb{I}_a)|\\&=-\lim_{N\to\infty}\frac{1}{\log N}\ln|\mathcal{L}(\beta,t)|,\end{aligned} \tag{15}$$

whose non-analyticities, signaling DQPTs, occur precisely at the nontrivial zeros of the zeta function. We further define the thermodynamic entropy density as

$$\begin{aligned}\mathcal{S}(\beta,t)&=-\frac{\partial}{\partial T}\left(\beta^{-1}\mathcal{F}_1(\beta,t)\right)\\&=-\mathcal{F}_1(\beta,t)+\beta\frac{\partial}{\partial\beta}\mathcal{F}_1(\beta,t),\end{aligned} \tag{16}$$

whose asymmetric jumps across the nontrivial zeros reveal the underlying non-equilibrium nature of the dynamical phase transitions (see Methods IV B for more details). For β slightly below a zero, the increase of $\mathscr{S}(\beta,t)$ corresponds to a loss of quantum coherence and an effective flow of information into internal degrees of freedom, consistent with thermalization. For β slightly above the zero, the decrease of $\mathscr{S}(\beta,t)$ reflects information backflow and partial re-coherence, signaling a revival of order. Thus, each nontrivial zero represents a non-equilibrium critical point at which the direction of information flow reverses—a dynamical analog of the critical line in the RH, detailed in Supplementary Note 3.

If the RH holds, such singularities occur exclusively along the critical line $\beta = 1/2$, analogous to Lee-Yang zeros in thermal systems[32–34]. For other temperatures $\beta \neq 1/2$, no such phenomenon arises. In this interpretation, the RH asserts that the system exhibits a unique non-equilibrium critical phase transition in the complex (β, t) plane, separating dynamically distinct regions of quantum evolution. According to equation (15), the same correspondence can also be realized in a pure-state setting, and we omit the detailed derivation here for brevity.

To access $\mathcal{L}(\beta,t)$ experimentally, we introduce a probe spin prepared in the superposition state $|+\rangle = (|\uparrow\rangle + |\downarrow\rangle)/\sqrt{2}$ and couple it to the system prepared in the thermal state ρ_s. Specifically, the $|\downarrow\rangle$ branch of the probe first evolves under the Hamiltonian

$$\mathcal{H}_{c1} = \lambda |\downarrow\rangle\langle\downarrow| \otimes (I - \sigma_z^1) \tag{17}$$

for a duration t_0, satisfying $\lambda t_0 = \pi/2$. This imprints a phase factor of $(-1)^{n+1}$ onto the n-th component $|n\rangle\langle n|$ of ρ_s. The branch then evolves under

$$\mathcal{H}_{c2} = |\downarrow\rangle\langle\downarrow| \otimes \mathcal{H}_0 \tag{18}$$

for a duration t, while the $|\uparrow\rangle$ branch remains unperturbed to serve as a reference. By measuring the probe coherence, we can directly extract

$$\langle\sigma_x\rangle + i\langle\sigma_y\rangle = \mathrm{Tr}\left[\rho_s e^{-i\mathcal{H}_0 t}\sigma_z^1\right] = \mathcal{L}(\beta,t), \tag{19}$$

thereby providing experimental access to the dynamical signatures.

Loschmidt echo corresponding to the RH

We further construct a complementary quantum system whose Loschmidt amplitude is directly linked to the zeta function and is better suited to probing high-order zeros. As in the first scheme, we begin with the physical Hamiltonian $\mathcal{H}_0^{\mathrm{ph}}$ with logarithmic spectrum E_n defined in Eq. (2), and adopt the same dimensionless variables as in Eq. (4), namely the inverse temperature β, the rescaled Hamiltonian $\mathcal{H}_0$, and the time t. The system is then prepared in the thermal equilibrium state ρ_s of $\mathcal{H}_0$ and coupled to a probe spin, forming the composite mixed state $\rho_0 = |\uparrow\rangle\langle\uparrow| \otimes \rho_s$.

The joint system then evolves for a duration t under the time-dependent Hamiltonian

$$\mathcal{H}_c(t') = \sigma_x \otimes \left(\mathcal{H}_0 - \dot{\theta}(t')I\right), \tag{20}$$

where $\theta(t)$ is the Riemann–Siegel theta function associated with $\zeta(1/2 + it)$, and the corresponding time-evolution operator is $U(t) = \exp[-i\int_0^t \mathcal{H}_c(t')\,dt']$.

The generalized Loschmidt amplitude (GLA) is defined as the overlap of the initial density matrix with the time-evolution operator[22],

$$\mathcal{G}(\beta,t) = \mathrm{Tr}[e^{-i\int_0^t \mathcal{H}_c dt'}\rho_0]. \tag{21}$$

Using the Riemann–Siegel approximation[35] for the Hardy Z-function $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2}+it)$, the GLA $\mathcal{G}\left(\frac{1}{2},t\right)$ can be written as

$$\mathcal{G}\left(\tfrac{1}{2},t\right) = \frac{1}{2\,\mathcal{Z}\left(\frac{1}{2},\mathcal{H}_0\right)}\left(e^{i\theta(t)}\sum_{n=1}^{N} n^{-\frac{1}{2}-it} + e^{-i\theta(t)}\sum_{n=1}^{N} n^{-\frac{1}{2}+it}\right) \xrightarrow[t\to\infty]{N=\sqrt{t/2\pi}} \frac{e^{i\theta(t)}\zeta\left(\frac{1}{2}+it\right)}{2\,\mathcal{Z}\left(\frac{1}{2},\mathcal{H}_0\right)}. \tag{22}$$

in the joint thermodynamic and long-time limit. The Loschmidt echo (LE) is defined as $\mathcal{B}(t) = |\mathcal{G}(t)|^2$, from which the Loschmidt rate (or dynamical free energy) is obtained as

$$\mathcal{F}_2(t) = -\lim_{N\to\infty}\frac{1}{\log N}\ln\mathcal{B}(t). \tag{23}$$

At $\beta = \frac{1}{2}$, the LE exhibits vanishing-and-revival dynamics, vanishing at the critical times t_c that correspond to the nontrivial zeros. These zeros of the LE generate non-analyticities in $\mathcal{F}_2(t)$, signaling DQPTs[20,22]. As the evolution time t increases, this correspondence becomes increasingly accurate. Taken together, the two systems fit into a unified framework based on a probe qubit coupled to a logarithmic spectrum, while differing in observables, dynamical protocols, and asymptotic limits.

We demonstrate that the GLA can be accessed via the expectation value of σ_z on the probe spin (see Methods IV C). Figure 2a shows numerical simulations of a three-spin system reproducing the zeros of the GLA corresponding to nontrivial zeros with imaginary parts in the range [420, 450]. Extending to a ten-spin system enables access to zeros with imaginary parts in $[6.595\times10^6, 6.595\times10^6 + 10]$, corresponding to the 13502344-th to 13502366-th zeros, as shown in Fig. 2b. An eighteen-spin system further captures the DQPTs linked to the 10^{12}-th and nearby zeros, as shown in Fig. 2c. Notably, the deviation between the estimated zeros and exact zeros decreases with increasing t, indicating improved precision of correspondence at larger zeros. In addition, we consider a pure initial state $\frac{1}{C}\sum_{n=1}^{N} n^{-\beta/2}|0\rangle|n\rangle$ evolving under the system Hamiltonian $\mathcal{H}_c$. This system likewise exhibits DQPTs, characterized by non-analytic behavior in the Loschmidt amplitude, whose zeros correspond to the nontrivial zeros of the zeta function.

Experimental verification using nuclear spins

The connection between physical quantum systems and the RH offers a novel method for experimentally probing the behavior of the zeta function through controlled Hamiltonian dynamics. As a proof-of-principle demonstration, we realized a five-qubit system on a nuclear magnetic resonance (NMR) platform. The qubits are encoded in the nuclear spins of a 1-bromo-2,4,5-trifluorobenzene molecule, partially aligned in the nematic liquid crystal solvent N-(4-methoxybenzaldehyde)-4-butylaniline (MBBA)[36–38]. The molecule consists of three ^{19}F nuclei and two ^{1}H nuclei. A single nuclear spin serves as the probe qubit, with the remaining spins constituting a sixteen-dimensional working system. The effective Hamiltonian of this five-qubit system in the rotating frame is given by

$$\mathcal{H}_{\mathrm{NMR}} = \sum_{j=0}^{4}\pi\nu_j\frac{\sigma_z^j}{2} + \sum_{j<k}\frac{\pi}{2}\left(J_{jk} + 2D_{jk}\right)\sigma_z^j\sigma_z^k, \tag{24}$$

where σ_z^j is the Pauli-z operator on the j-th spin, ν_j denotes the chemical shift of the j-th spin, and $(J_{jk} + 2D_{jk})$ represents the effective coupling strength between spins j and k. Here, J_{jk} and D_{jk} correspond to the scalar and residual dipolar couplings, respectively. The molecular structure and Hamiltonian parameters are illustrated in Fig. 3a, b, respectively. All experiments were performed on a Bruker Avance III 600 MHz spectrometer at $T = 305$ K.

There are two major challenges in experimentally realizing this constructed Hamiltonian dynamics. The first is the preparation of

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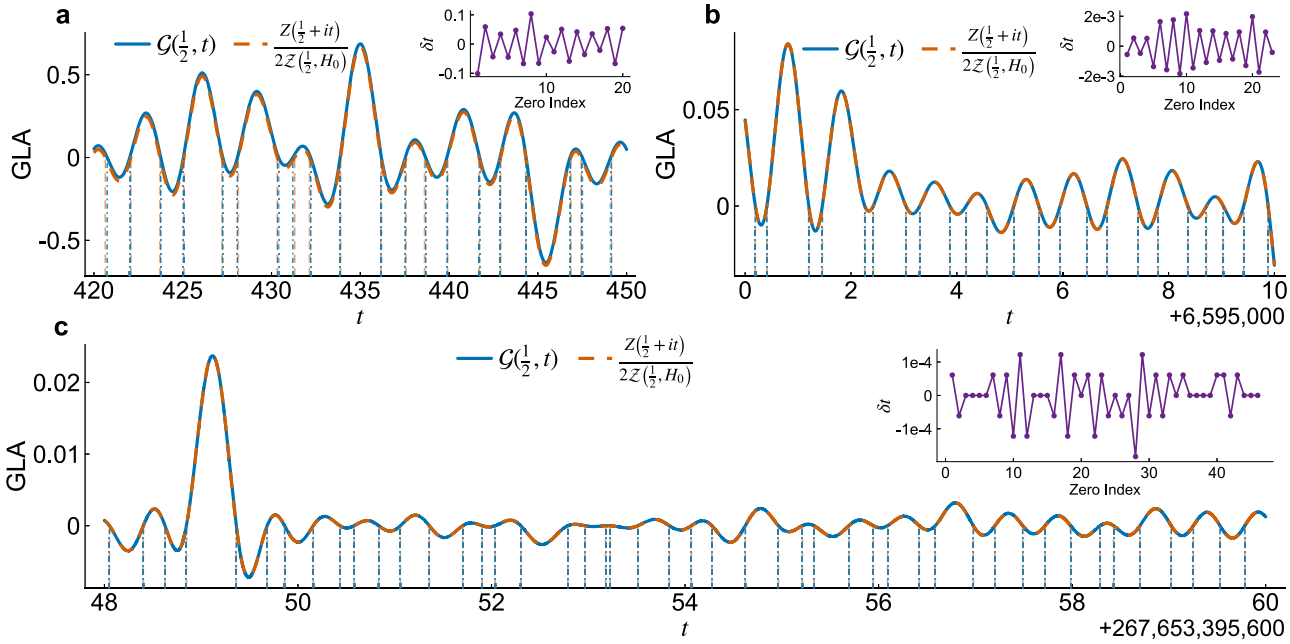


Fig. 2 | Generalized Loschmidt amplitude $\mathcal{G}(\frac{1}{2}, t)$ across different imaginary part regions. a Nontrivial zeros with imaginary parts in the range [420, 450], corresponding to the 216th to 235th zeros, obtained using a 3-spin quantum system. **b** Nontrivial zeros with imaginary parts in the range [6.595×10^6, $6.595 \times 10^6 + 10$], corresponding to the 13502344-th to 13502366-th zeros, obtained using a 10-spin quantum system. **c** Nontrivial zeros with imaginary parts in the range [267653395648, 267653395660], corresponding to the $(10^{12} - 3)$-th to $(10^{12} + 43)$-th zeros, obtained using an 18-spin quantum system. The blue solid line shows the simulated $\mathcal{G}(\frac{1}{2}, t)$, while the red dashed line shows $Z(\frac{1}{2} + it)/(2\mathcal{Z}(\frac{1}{2}, \mathcal{H}_0))$. Vertical red dashed lines mark the exact zeros, and blue dashed and dotted lines mark the zeros estimated by the quantum system. Insets show the deviation δt between the exact zeros and the estimated zeros, which decreases with increasing t. The zero-index annotations are assigned using tabulated zeta-zero data from LMFDB for panels (**a**, **b**) and from Odlyzko's tabulation for panel (**c**)[3].

thermal states, and the second is the implementation of controlled dynamical evolution. NMR systems are ensemble-based and naturally begin in a thermal equilibrium state

$$\rho_{\mathrm{eq}} \sim \gamma_{\mathrm{F}} \sum_{i=1}^{3} \sigma_z^i + \gamma_{\mathrm{H}} \sum_{i=4}^{5} \sigma_z^i, \tag{25}$$

where we ignore the identity component for convenience and γ_{F} and γ_{H} denote the gyromagnetic ratios of the fluorine and hydrogen nuclei, respectively. Starting from the initial state ρ_{eq}, we can redistribute the level populations via coherent control and subsequently eliminate the off-diagonal elements to arrive at a state close to the target

$$\rho_0(\beta) \sim \sigma_z \otimes \sum_{n=1}^{16} n^{-\beta} |n\rangle\langle n|. \tag{26}$$

Figure 3c presents our procedure for preparing the thermal state (see Methods IV A for more details).

In our experiment, two controlled Hamiltonian dynamics under $\mathcal{H}_{\mathrm{c1}}$ and $\mathcal{H}_{\mathrm{c2}}$ are implemented. The first $\mathrm{e}^{-i\mathcal{H}_{\mathrm{c1}} t_0}$ corresponds to a controlled-Z gate between the probe spin and the last spin. The second $\mathrm{e}^{-i\mathcal{H}_{\mathrm{c2}} t}$ can be decomposed into multiple multi-qubit controlled-$R_z(\theta_n)$ operations. The rotation angles are given by $\theta_n = t \ln n$, where t denotes the imaginary part of s. Finally, we measure the probe-spin coherence through the element

$$\mathcal{L}(\beta, t) = \langle \uparrow | U_{\mathrm{tot}} \rho_0 U_{\mathrm{tot}}^{\dagger} | \downarrow \rangle, \tag{27}$$

where the total evolution operator is

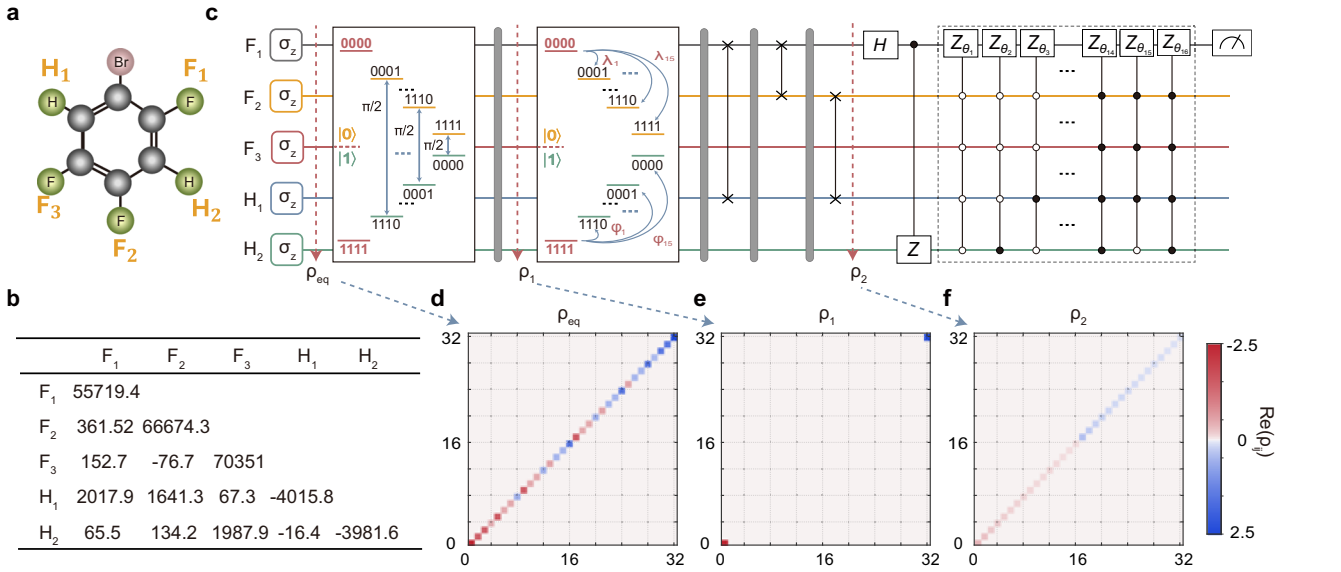
$$U_{\mathrm{tot}} = \mathrm{e}^{-i\mathcal{H}_{\mathrm{c2}} t} \mathrm{e}^{-i\mathcal{H}_{\mathrm{c1}} \pi/(2\lambda)} (H \otimes I). \tag{28}$$

Figure 3 illustrates the full experimental sequence, from initial state preparation to final measurement. To enhance control accuracy, we implement all coherent operations and unitary evolutions using shaped pulses optimized via a gradient ascent pulse technique[39], yielding a numerical simulation fidelity exceeding 99.5%.

We conducted three types of experiments to comprehensively validate the theoretical predictions. First, we prepare the thermal state $\rho_0(\beta)$ with $\beta = 0.5$ (corresponding to the RH) and vary the evolution time t under the Hamiltonian $\mathcal{H}_{\mathrm{c2}}$. Second, we prepare another thermal state $\rho_0(\beta)$ with $\beta = 0.3$, which lies off the critical line of the RH, and again vary the evolution time t. Third, we fix the evolution time to the imaginary part of the first nontrivial zero, and scan β over the range [0.1, 0.9]. Then the coherence dynamics $\mathcal{L}(\beta, t)$ of the probe qubit is measured in each configuration. Figure 4a, b present the coherence dynamics $\mathcal{L}(\beta, t)$ under different values of β. When $\beta = 0.5$, the coherence of the probe spin exhibits recurring vanishing-and-revival behavior, and the points where both the real and imaginary parts of $\mathcal{L}(\beta, t)$ approach zero correspond to nontrivial zeros of the zeta function. As a demonstration, coherence zeros at $t_1 = 14.12$ ($t_1^{\mathrm{th}} = 14.13$), $t_2 = 20.96$ ($t_2^{\mathrm{th}} = 21.02$), $t_3 = 25.09$ ($t_3^{\mathrm{th}} = 25.01$), $t_4 = 30.44$ ($t_4^{\mathrm{th}} = 30.43$), and $t_5 = 32.93$ ($t_5^{\mathrm{th}} = 32.94$) are extracted through polynomial fitting of the experimental data and agree well with the ideal zeros of the zeta function. In contrast, for $\beta = 0.3$, the coherence dynamics show no discernible vanishing-and-revival behavior, and no significant coherence zeros are observed.

The third configuration, scanning β at fixed t, is shown in Fig. 4c, where $\mathcal{L}(\beta, t)$ exhibits a clear minimum at $\beta = 0.5$, confirming that the zero-crossing behavior is uniquely tied to the critical line. Figure 5 illustrates the behavior of the free-energy density as a function of β and t. As the system size increases, coherence zeros and DQPTs occur upon crossing the critical line $\beta = 0.5$, in accordance with the RH. We also observe a regularly spaced sequence of singularities on the line $\beta = 1$,

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	F_1	F_2	F_3	H_1	H_2
F_1	55719.4				
F_2	361.52	66674.3			
F_3	152.7	-76.7	70351		
H_1	2017.9	1641.3	67.3	-4015.8	
H_2	65.5	134.2	1987.9	-16.4	-3981.6

Fig. 3 | Physical system and quantum circuit for simulating the Riemann zeta function. a Molecular structure of the 1-bromo-2,4,5-trifluorobenzene. The five nuclear spins are encoded as qubits, with the F_1 spin serving as the probe qubit to extract the average accumulated phase factor. **b** Measured Hamiltonian parameters. The chemical shifts ν_i are shown on the diagonal, while the off-diagonal elements represent the coupling strengths ($J_{jk} + 2D_{jk}$), both in units of Hz. **c** Quantum circuit implementation, including thermal state preparation, controlled dynamical evolution, and coherence measurement. H denotes the Hadamard gate acting on the probe qubit. Coherent operations for state preparation and multi-qubit quantum gates are realized using shaped control pulses. The parameters λ_i and φ_i for population transfer are determined by β, and the parameters θ_i are determined by t. **d**–**f** Effective population evolution during state preparation. Here, we employ the traceless deviation density matrix to describe the state evolution.

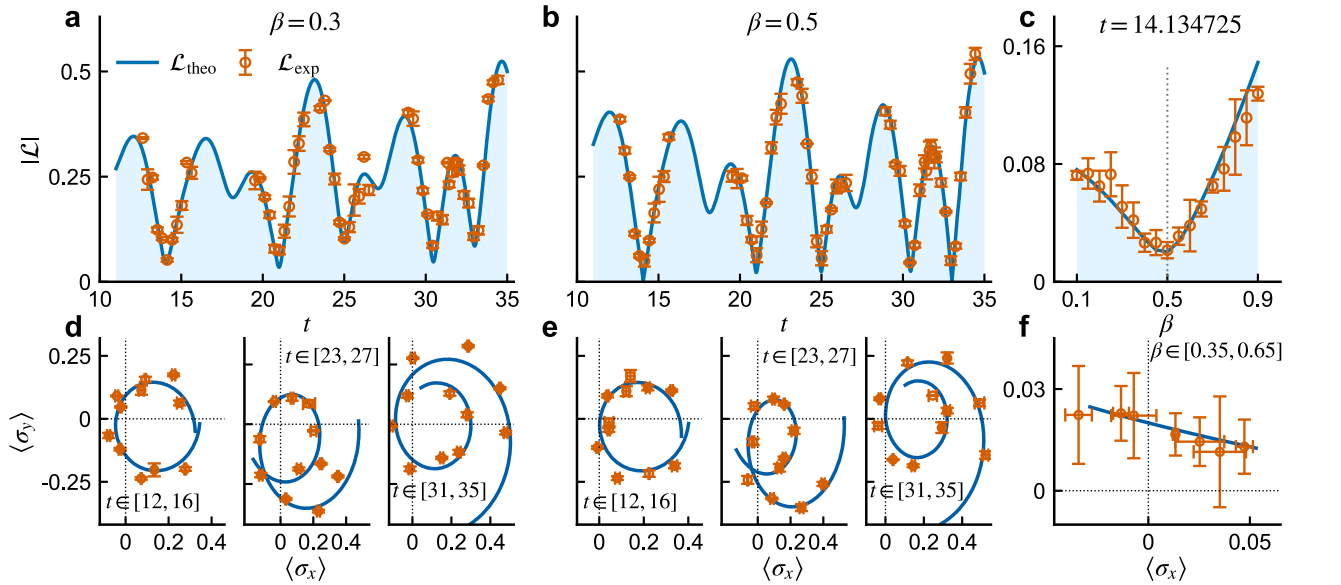


Fig. 4 | The coherence dynamics $\mathcal{L}(\beta, t)$ of the probe qubit and the nontrivial zeros. a, **b** Measured probe-qubit coherence $\mathcal{L}(\beta, t)$ as a function of evolution time t for $\beta = 0.3$ (**a**) and $\beta = 0.5$ (**b**). **c** Measured probe-qubit coherence $|\mathcal{L}(\beta, t)|$ as a function of β at fixed time $t = 14.134725$, equal to the imaginary part of the first nontrivial zero. Solid lines and open circles denote numerically calculated and experimental values of the probe spin coherence, respectively. **d**, **e** Behavior of the coherence as a function of t for fixed $\beta = 0.3$ (**d**) and $\beta = 0.5$ (**e**). The first, third and fifth nontrivial zeros are identified at times where both the real part $\langle\sigma_x\rangle$ and the imaginary part $\langle\sigma_y\rangle$ vanish simultaneously. For $\beta = 0.5$, which lies on the critical line of the RH, the probe-qubit coherence exhibits multiple zeros that match the nontrivial zeros one-to-one, whereas for $\beta = 0.3$, away from the critical line, no such zeros are observed. **f** The behavior of the coherence for $\beta \in [0.35, 0.65]$ at a fixed t corresponding to the first nontrivial zero of the zeta function, showing that the coherence zero is approached near $\beta = 0.5$. The circles show the mean, and the error bars denote one standard deviation of the measured values from repeated experiments in this figure.

arising from the poles of $1/(1 - 2^{1-s})$ and reflecting the inapplicability of the Dirichlet-series representation along $\beta = 1$.

Thus, the non-analyticity of $\mathcal{F}_1(\beta, t)$ at nontrivial zeros provides a physical manifestation of the Riemann Hypothesis: the hypothesis is equivalent to the statement that the dynamical free-energy density of a suitably constructed quantum system becomes singular only along the critical line $\beta = 0.5$ in the complex temperature-time plane within $0 < \beta < 1$.

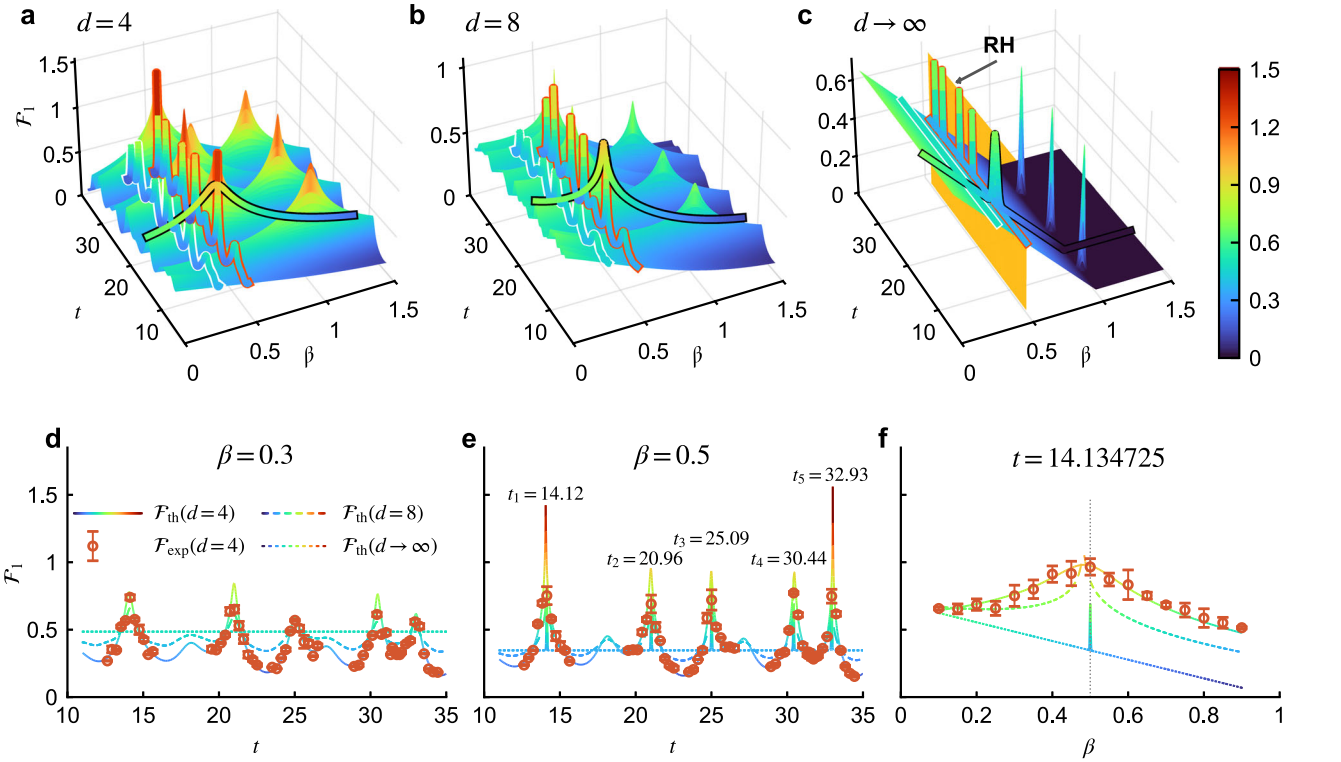


Fig. 5 | Free-energy density $\mathcal{F}_1(\beta, t)$ as a function of β and t. a–c Theoretical $\mathcal{F}_1(\beta, t)$ for different system sizes $d = \log N$ (**a,** $d = 4$; **b,** $d = 8$; **c,** $d \to \infty$). In the thermodynamic limit, $\mathcal{F}_1(\beta, t)$ exhibits discontinuous jumps from $(1-\beta)\ln 2$ to $\ln 2$ at $\beta = 1/2$, signaling DQPTs. **d–f** Experimental signatures of the non-analytic behavior in the free-energy density. Error bars correspond to one standard deviation from repeated experiments. **d, e** Cuts of $\mathcal{F}_1(\beta, t)$ as a function of t at fixed $\beta = 0.3$ (**d**) and $\beta = 0.5$ (**e**), comparing theoretical predictions (th) for different d with experimental data (exp). In **e**, critical times are marked at $t_1 = 14.12$, $t_2 = 20.96$, $t_3 = 25.09$, $t_4 = 30.44$ and $t_5 = 32.93$, whereas no such behavior is observed in (**d**), consistent with the RH. **f** $\mathcal{F}_1(\beta, t)$ as a function of β at fixed $t = 14.134725$, corresponding to the imaginary part of the first nontrivial zero.

Gate-based quantum computing framework for probing the RH

Since the two dynamical constructions introduced above admit many-body spin representations with exponentially many interaction terms, their direct realization on a gate-based quantum computer is highly nontrivial[40]. Therefore, we propose an efficient computational framework that enables their realization on a universal gate-based quantum computer. The constructions in Sections II A and II B can be reduced to two fundamental tasks:

(i) Preparation of the initial thermal state, which encodes the real part of the zeta argument.
(ii) Simulation of time evolution under the target Hamiltonian, which encodes the imaginary part.

According to the Choi-Jamiołkowski isomorphism, the initial thermal state can be encoded as a pure state $|\psi_0\rangle = \frac{1}{C}\sum_{n=1}^{N} n^{-\beta/2}|n\rangle$. It can be prepared to precision ε using a polynomial number of qubits and basic gates. We further show that the evolution $U(t) = \mathrm{e}^{-i\mathcal{H}_0 t}$ for $\mathcal{H}_0 = \sum_{n=1}^{N} \ln n|n\rangle\langle n|$ can also be implemented to precision ξ with polynomial resources, and controlled evolutions incur the same polynomial overhead. These guarantees ensure scalability of our method on a gate-based quantum computer (see Methods IV D and Fig. 6 for details).

We then apply this framework to accelerate the verification and probing of nontrivial zeros via the Riemann–Siegel formula, which decomposes the zeta function into two partial summations that can be evaluated using systems similar to those in Section II A. To achieve a total error δ in the zeta function, the quantum circuit complexities are bounded by

$$\mathrm{Poly}(\log \delta^{-1}, \log|t|, \log(1-\beta)^{-1}, \log\beta^{-1}). \tag{29}$$

The corresponding sample complexities for the two summations are bounded by

$$\begin{aligned} R_{s1} &= \mathcal{O}(\delta^{-1})(1-\beta)^{-1}|t|^{(1-\beta)/2}, \\ R_{s2} &= \mathcal{O}(\delta^{-1})\beta^{-1}|t|^{(1-\beta)/2}. \end{aligned} \tag{30}$$

Hence, the overall computational complexity (See Methods IV E for details) is bounded by

$$\delta^{-1}|t|^{(1-\beta)/2}\mathrm{Poly}(\log\delta^{-1}, \log|t|, (1-\beta)^{-1}, \beta^{-1}). \tag{31}$$

Specializing to locating and verifying nontrivial zeros, the known explicit zero-free region[41] implies that the relevant range is

$$\frac{1}{2} \le \beta \le 1 - \frac{1}{\mathcal{O}\left((\ln|t|)^{2/3}(\ln\ln|t|)^{1/3}\right)}. \tag{32}$$

In this regime, the complexity bound simplifies to

$$\delta^{-1}|t|^{(1-\beta)/2}\mathrm{Poly}(\log\delta^{-1}, \log|t|). \tag{33}$$

Thus, the $|t|$-dependence reduces from $\sqrt{|t|}$ (direct Riemann-Siegel evaluation) to $|t|^{(1-\beta)/2}$ (at most $|t|^{1/4}$), yielding at least a quadratic speedup. Table 1 summarizes these complexity bounds.

Discussion

In this work, we have established a direct correspondence between the RH and the dynamics of engineered quantum many-body systems, and have experimentally verified this connection on a physical platform. The mapping between nontrivial zeros and DQPTs reveals a deep and

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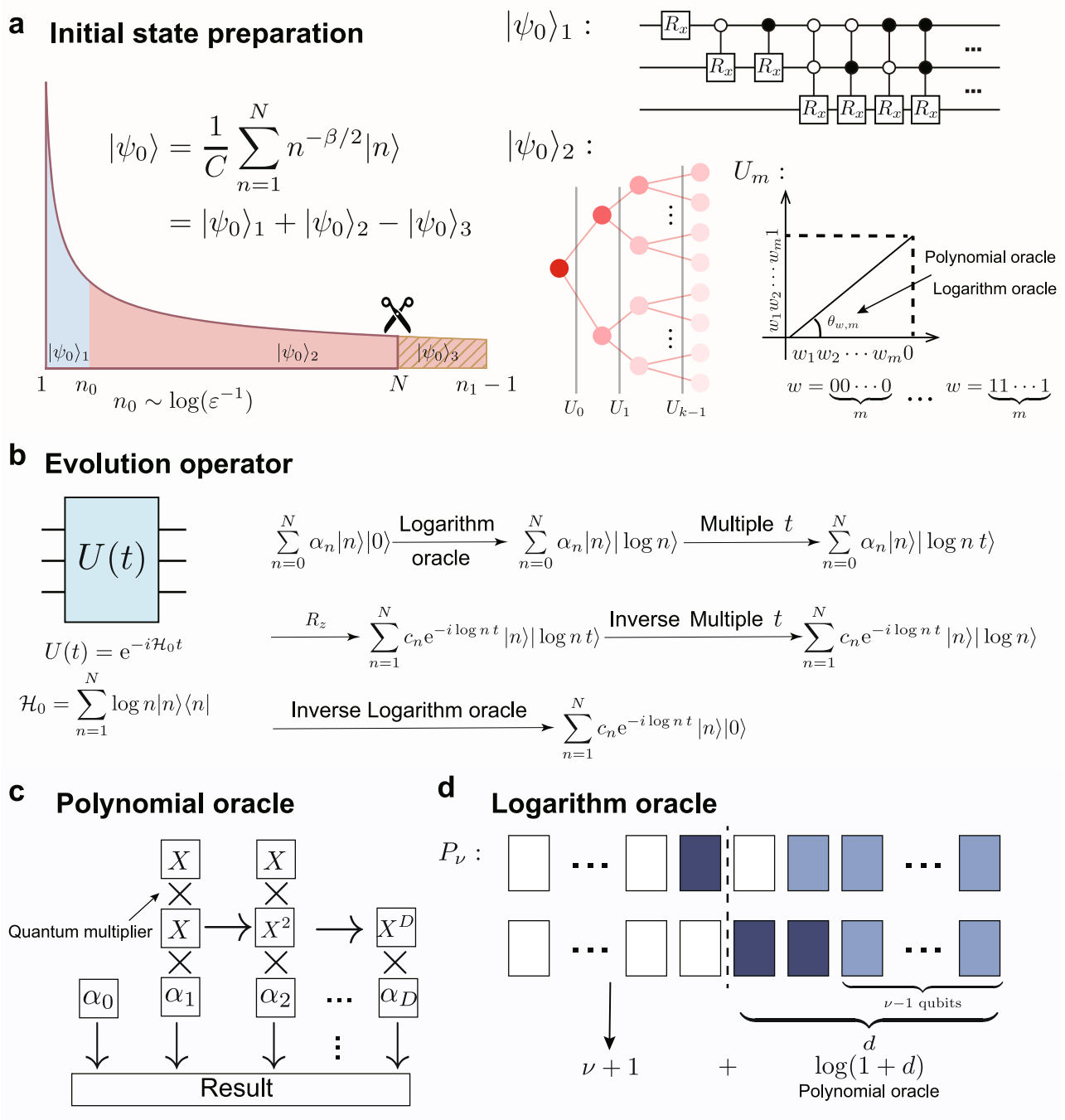


Fig. 6 | Construction of the quantum system on a gate-based quantum computer. a The initial state is prepared as $|\psi_0\rangle = |\psi_0\rangle_1 + |\psi_0\rangle_2 - |\psi_0\rangle_3$ (normalization omitted). The components $|\psi_0\rangle_1$, $|\psi_0\rangle_2$, and $|\psi_0\rangle_3$ correspond to the blue, orange, and slashed yellow regions, respectively. The states $|\psi_0\rangle_1$ and $|\psi_0\rangle_2$ are generated using controlled rotations and Grover's amplitude-splitting method, respectively. **b** The time-evolution operator $U(t) = e^{-i\mathcal{H}_0 t}$ is implemented in five steps: logarithm computation, multiplication by t, phase imprinting, inverse multiplication, and inverse logarithm computation. **c**, **d** The polynomial and logarithmic oracles serve as preliminary building blocks for implementing the time-evolution operator and preparing the initial state.

unexplored bridge between number theory and quantum physics. In particular, we have demonstrated that a thermally equilibrated quantum system, when driven by a tailored Hamiltonian, undergoes time-domain phase transitions whose critical points align precisely with the nontrivial zeros. To verify large nontrivial zeros, we also construct a quantum system that undergoes time-domain phase transitions whose critical points align increasingly accurately with the large nontrivial zeros. Together, these results suggest that the mysterious structure of the zeta function may admit a physical interpretation in terms of quantum critical dynamics. Conversely, the DQPTs triggered by the nontrivial zeros exhibit extreme sensitivity, enabling high-precision measurements of temperature and time.

Although the logarithmic spectrum $E_n = \ln n$ does not occur in natural condensed-matter systems, it forms the foundation of our two engineered quantum many-body constructions. We demonstrate that both settings can be efficiently implemented on universal quantum hardware with polynomial time complexity, and numerical verification of the RH can be achieved by our method with at least a quadratic speedup over classical methods. Beyond this, our results suggest that nontrivial zeros also encode signatures of information propagation in

Table 1 | Complexity bounds for evaluating $\zeta(\beta + it)$ with precision δ across different regions using our quantum algorithm, compared to classical approaches

Region of β	Sample complexity $\mathcal{R}_s$	Total complexity $\mathcal{R}_c\mathcal{R}_s$
$0 < \beta < 1$	$\mathcal{O}(\delta^{-1})\lvert t\rvert^{(1-\beta)/2}(\beta^{-1}+(1-\beta)^{-1})$	$\delta^{-1}\lvert t\rvert^{(1-\beta)/2}\mathrm{Poly}(\log\delta^{-1},\ \log\lvert t\rvert, (1-\beta)^{-1}, \beta^{-1})$
Possible zero region	$\mathcal{O}(\delta^{-1})\lvert t\rvert^{(1-\beta)/2}(\beta^{-1}+(1-\beta)^{-1})$	$\delta^{-1}\lvert t\rvert^{(1-\beta)/2}\mathrm{Poly}(\log\delta^{-1},\ \log\lvert t\rvert)$
$\beta=\frac{1}{2}$	$\mathcal{O}(\delta^{-1})\lvert t\rvert^{1/4}$	$\delta^{-1}\lvert t\rvert^{1/4}\mathrm{Poly}(\log\delta^{-1},\ \log\lvert t\rvert)$
Classical method	~ $\lvert t\rvert$ (Alternating series)[50]	
	~ $\lvert t\rvert^{1/2}$ (Riemann–Siegel formula)[35]	

The possible zero region, defined in Eq. (87), is the regime of interest for locating and verifying nontrivial zeros.

quantum systems, motivating future studies of their relationship to out-of-time-ordered correlators (OTOCs)[42–44] and other diagnostics of quantum chaos.

Given the inherent difficulty of verifying the RH using classical computation, our approach opens a path toward leveraging quantum computing as an algorithmic tool for probing one of mathematics' most enduring open problems. We note that more efficient classical methods for evaluating the zeta function on the critical line $\beta=\frac{1}{2}$ are known[3,45]. Our ongoing work explores how the proposed framework can accelerate the most time-consuming subroutines of these methods, thereby further improving their overall efficiency.

Moreover, the framework introduced here naturally extends to other series and special functions, including Dirichlet L-functions, and thus defines a broader class of number-theoretic benchmarks for demonstrating quantum advantage[46]. Since the prime number distribution can be expressed in terms of the zeta function and related L-functions, the same ideas may be adapted to quantum simulations of prime-counting observables and to the evaluation of quantities such as the Witten index in string and supersymmetric field theories.

Methods

Thermal state preparation in experiment

The preparation sequence, shown in the main text, consists of two analytically designed line-selective shaped pulses, four pulsed field gradients, and three SWAP gates. The first shaped pulse selectively depletes population from all states except $|\uparrow\uparrow\uparrow\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\downarrow\downarrow\downarrow\rangle$[38], which serve as logical sources for encoding the thermal-equivalent distribution. A gradient pulse is applied immediately afterward to suppress non-diagonal coherence terms generated in this step. The second shaped pulse redistributes the populations from $|\uparrow\uparrow\uparrow\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\downarrow\downarrow\downarrow\rangle$ into a target position matching the desired thermal distribution. This step introduces zeroth-order coherences that cannot be fully removed by gradients alone. To address this, three SWAP operations are inserted before the final two gradient pulses, transferring coherence terms from homonuclear fluorine spins to heteronuclear channels, where they can be completely dephased. Below, we detail the function of each component in the preparation protocol.

The thermal equilibrium state of this five-qubit system is highly mixed. It can be described by a deviation density matrix of the form

$$\rho_{\mathrm{eq}} \propto \gamma_F(\sigma_z^1+\sigma_z^2+\sigma_z^3)+\gamma_H(\sigma_z^4+\sigma_z^5), \tag{34}$$

where γ_H and γ_F are the gyromagnetic ratios for ^{1}H and ^{19}F nuclei, and σ_z^i is the Pauli-z operator for the i-th qubit. This matrix is traceless and diagonal, with the maximal populations residing in the $|\uparrow\uparrow\uparrow\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\downarrow\downarrow\downarrow\rangle$ states. Furthermore, pairs of symmetric basis states (e.g., $|\uparrow\downarrow\uparrow\downarrow\uparrow\rangle$ and $|\downarrow\uparrow\downarrow\uparrow\downarrow\rangle$) possess populations of equal magnitude but opposite sign, satisfying $P_{|\uparrow\downarrow\uparrow\downarrow\uparrow\rangle}+P_{|\downarrow\uparrow\downarrow\uparrow\downarrow\rangle}=0$. This intrinsic symmetry is key to our line-selective approach, allowing us to manipulate populations.

The first shaped pulse is designed to selectively deplete the population of all energy levels except for the $|\uparrow\uparrow\uparrow\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\downarrow\downarrow\downarrow\rangle$ states[38], which serve as the source states for subsequent redistribution. This is achieved by applying a series of analytically constructed $\pi/2$ line-selective rotations between all pairs of symmetric basis states (e.g., $|\uparrow\downarrow\uparrow\downarrow\uparrow\rangle$ and $|\downarrow\uparrow\downarrow\uparrow\downarrow\rangle$). For a given pair of states, $|x\rangle$ and $|\bar{x}\rangle$, the initial population imbalance is transformed by an $R_y(\pi/2)$ rotation as

$$\begin{pmatrix} a & 0 \\ 0 & -a \end{pmatrix} \xrightarrow{R_y(\pi/2)} \begin{pmatrix} 0 & a \\ a & 0 \end{pmatrix}. \tag{35}$$

Since these pairs have opposite initial populations in ρ_{eq}, applying a $\pi/2$ rotation about the y-axis effectively converts the diagonal population matrix to a purely off-diagonal one, which is then removed by a subsequent gradient pulse along the z-direction. Because the system has an odd number of qubits, this procedure avoids generating zeroth-order coherences. A pulsed field gradient is then applied to remove any residual higher-order coherences, leading to a diagonal intermediate state ρ_1 in which only the $|\uparrow\uparrow\uparrow\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\downarrow\downarrow\downarrow\rangle$ populations remain.

The second shaped pulse redistributes the populations of $|\uparrow\uparrow\uparrow\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\downarrow\downarrow\downarrow\rangle$ into a coherent superposition that encodes the target thermal-equivalent state. This state is represented by

$$\rho=\sigma_z\otimes\sum_{n=1}^{16}\frac{n^{-\beta}}{\mathcal{N}^2}|n\rangle\langle n|, \tag{36}$$

with the normalization constant $\mathcal{N}$. This is accomplished by a sequence of analytically determined line-selective rotations that couple the source basis to the remaining basis. Each rotation is precisely calculated to transfer a fraction of the population from $|\uparrow\uparrow\uparrow\uparrow\uparrow\rangle$ or $|\downarrow\downarrow\downarrow\downarrow\downarrow\rangle$ to a selected target basis. The rotation angles are derived from

$$\theta_k=2\sin^{-1}\left(\sqrt{p_k/P_{\mathrm{rem}}}\right), \tag{37}$$

where p_k is the target population of the k-th basis and P_{rem} is the remaining unallocated population. This asymmetric redistribution process inevitably introduces zeroth-order coherences, particularly among homonuclear fluorine spins, which cannot be effectively suppressed by gradient fields alone.

To address the issue of residual homonuclear coherences, we employ a SWAP-assisted gradient scheme. The core idea is to transfer these coherences from homonuclear channels to heteronuclear ones, where they can be completely dephased by a gradient pulse. The sequence begins with a SWAP(1,4) gate to exchange the states of qubit 1 (^{19}F) and qubit 4 (^{1}H). This operation converts fluorine-fluorine coherences (e.g., between qubits 1 and 2) into heteronuclear coherences that can be dephased by the first gradient pulse. A second SWAP(1,4) gate restores the original logical labeling. To eliminate the remaining fluorine-fluorine coherence between qubits 2 and 3, we next apply a SWAP(2,4) gate, shifting this coherence to the heteronuclear (3,4) channel, which is then removed by the second gradient pulse. For increased efficiency, the two middle SWAP gates, SWAP(1,4) and

SWAP(2,4), can be combined into a single SWAP(1,2) operation. A final SWAP(2,4) then returns the system to its original qubit configuration. This sequence of SWAP gates and gradient pulses ensures that all zeroth-order coherences are effectively eliminated, yielding a clean thermal-equivalent state with a numerical fidelity exceeding 99%.

Free-energy and entropy density singularities

The free-energy density is defined as

$$\mathcal{F}_1(\beta,t)=\lim_{N\to\infty}-\frac{1}{d}\ln|\mathcal{L}(\beta,t)|, \tag{38}$$

where $d=\log N$ denotes the number of degrees of freedom. According to equation (10), the accumulated phase factor can be expressed as

$$\mathcal{L}(\beta,t)=S_N(s)/\mathcal{Z}(\beta,\mathcal{H}_0), \tag{39}$$

where $S_N(s)=\sum_{n=1}^{N}(-1)^{n+1}n^{-s}$. We prove that (see Supplementary Notes 1 and 2)

$$\lim_{N\to\infty}-\frac{1}{d}\ln(|S_N(s)|)=\begin{cases}\beta\ln 2, & \zeta(s)=0,\\ 0, & \text{Otherwise},\end{cases} \tag{40}$$

and

$$\lim_{N\to\infty}-\frac{1}{d}\ln\left(\mathcal{Z}(\beta,\mathcal{H}_0)\right)=\begin{cases}(\beta-1)\ln 2, & 0<\beta<1,\\ 0, & \beta>1.\end{cases} \tag{41}$$

Combining these results, the free-energy density in the thermodynamic limit is given by

$$\begin{aligned}&\mathcal{F}_1(\beta,t)\\&=\lim_{N\to\infty}-\frac{1}{d}\left(\ln(|S_N(s)|)-\ln\left(\mathcal{Z}(\beta,\mathcal{H}_0)\right)\right)\\&=\begin{cases}\ln 2, & \zeta(s)=0,\\ (1-\beta)\ln 2, & 0<\beta<1,\ \zeta(s)\neq 0,\\ 0, & \beta>1.\end{cases}\end{aligned} \tag{42}$$

Therefore, the thermodynamic entropy density is given by

$$\begin{aligned}\mathcal{S}(\beta,t)&=-\mathcal{F}_1(\beta,t)+\beta\frac{\partial}{\partial\beta}\mathcal{F}_1(\beta,t)\\&=\begin{cases}-\ln 2, & 0<\beta<1,\ \zeta(s)\neq 0,\\ 0, & \beta>1.\end{cases}\end{aligned} \tag{43}$$

in the thermodynamic limit. At the nontrivial zeros of the zeta function, the free-energy density is nonanalytic, and the entropy density undergoes discontinuous jumps, providing strong evidence for the occurrence of DQPTs.

Measurement of the Loschmidt amplitude

We consider a system, similar to the one described in Section II B, designed to measure the Loschmidt amplitude defined in Eq. (22). The composite system is initialized in the same state $\rho_0=|\uparrow\rangle\langle\uparrow|\otimes\rho_s$, but now evolves under the coupling Hamiltonian $\mathcal{H}_{c2}=\frac{1}{2}\sigma_x\otimes(\mathcal{H}_0-\theta(t)\mathcal{I})$, which is one half of the Hamiltonian $\mathcal{H}_c$. The corresponding time-evolution operator is therefore $U(t)=\exp(-i\int_0^t\mathcal{H}_{c2}dt')=\exp(-\frac{i}{2}\int_0^t\mathcal{H}_c dt')$. Defining $A(t)=\mathcal{H}_0 t-\theta(t)I$, the density matrix after evolution is

$$\begin{aligned}U\rho_0U^\dagger=\frac{1}{2}\big[&|+\rangle\langle+|\otimes\rho_s+e^{-iA(t)}|+\rangle\langle-|\otimes\rho_s\\&+e^{iA(t)}|-\rangle\langle+|\otimes\rho_s+|-\rangle\langle-|\otimes\rho_s\big],\end{aligned} \tag{44}$$

where $|+\rangle=\frac{1}{\sqrt{2}}(|\uparrow\rangle+|\downarrow\rangle)$ and $|-\rangle=\frac{1}{\sqrt{2}}(|\uparrow\rangle-|\downarrow\rangle)$. The reduced density matrix of the probe qubit is

$$\rho_{\text{probe}}=\frac{1}{2}I+\frac{1}{2}\text{Re}S(t)\sigma_z+\frac{1}{2}\text{Im}\,S(t)\sigma_y, \tag{45}$$

in which

$$S(t)=\text{Tr}[e^{-iA(t)}\rho_s]=\frac{1}{\mathcal{Z}(\frac{1}{2},\mathcal{H}_0)}e^{i\theta(t)}\sum_{n=1}^{N}n^{-1/2-it}. \tag{46}$$

Consequently, measuring $\langle\sigma_z\rangle$ on the first qubit yields $\text{Re}S(t)=\mathcal{G}(\frac{1}{2},t)$.

Gate-based realization of quantum systems

In this subsection, we describe the construction of the target initial state

$$|\psi_0\rangle=\frac{1}{C}\sum_{n=1}^{N}n^{-\beta/2}|n\rangle \tag{47}$$

and the evolution operator

$$U(t)=e^{-i\mathcal{H}_0 t},\qquad \mathcal{H}_0=\sum_{n=1}^{N}\ln(n)|n\rangle\langle n| \tag{48}$$

on a standard gate-based quantum computer as shown in Fig. 6. We begin with two preliminary lemmas. The first establishes a quantum oracle for evaluating polynomial functions, while the second provides a procedure for approximating logarithmic functions to the required precision. Together, these results serve as building blocks for our main results: a method for constructing the initial state (Theorem 3) and an efficient implementation of the evolution operator (Theorem 4).

Lemma 1. (Polynomial oracle) Let $O(f)$ denote an oracle implementing the transformation

$$\sum_x\alpha_x|x\rangle|0\rangle\mapsto\sum_x\alpha_x|x\rangle\,|f(x)\rangle, \tag{49}$$

where $f(x)$ is a polynomial of degree at most D. Suppose the coefficients of f and the input x are specified to a_1 and a_2 bits of precision, respectively. The output is encoded with r_1 integer qubits and r_2 fractional qubits. Then $O(f)$ can be implemented using $\mathcal{O}(D^2a_2^2+D^2a_1a_2+D(r_1+r_2))$ gates and $\mathcal{O}(D^2a_2+a_1)$ ancilla qubits.

Proof sketch. Use reversible fixed-point arithmetic to generate the power chain $x, x^2, \ldots, x^D$. A reusable product register computes each term c_dx^d, adds it in place to the output, and is then uncomputed. Finally, reversing the power-generation circuit cleans the power workspace (see Supplementary Note 4 for details).

Lemma 2. (Logarithm oracle) Let L denote an oracle implementing the transformation

$$\sum_{n=1}^{N}\alpha_n|n\rangle|0\rangle\mapsto\sum_{n=1}^{N}\alpha_n|n\rangle|\widetilde{\log n}\rangle, \tag{50}$$

where $\widetilde{\log n}$ approximates $\log n$ to within error η. Then L can be implemented using

$$\mathcal{O}\left((\log N)^3\log^2(1/\eta)+(\log N)^2\log^3(1/\eta)\right) \tag{51}$$

gates and $\mathcal{O}(\log N\log^2(1/\eta))$ ancilla qubits.

Proof sketch. The construction proceeds in two steps. First, a register of ancilla qubits identifies the interval

$$P_\nu:\ 3\cdot 2^{\nu-1}\le n<3\cdot 2^{\nu},\ 1\le\nu\le\left\lceil\log\left(\frac{N+1}{3}\right)\right\rceil, \tag{52}$$

yielding a contribution $(\nu+1)$ in $\log n$. Second, approximate the residual contribution

$$\log(1+d),\qquad d=n/2^{\nu+1}-1, \tag{53}$$

to accuracy η using the polynomial oracle in Lemma 1. The result is encoded using $\log\log N$ qubits for the integer part and $\log(1/\eta)$ for the fractional part. Full details of the polynomial expansions and resource counts are provided in Supplementary Note 4.

Construction of the initial state. The target initial state can be written (formally, up to normalization and coefficients) as

$$|\psi_0\rangle=|\psi_0\rangle_1+|\psi_0\rangle_2-|\psi_0\rangle_3. \tag{54}$$

We first construct $|\psi_0\rangle_1$ using the Angle-preparation oracle U_m, and then recover the initial state via the linear combination of Unitaries (LCU) method and fixed-point amplitude amplification. A complete proof is provided in Supplementary Note 5.

Theorem 1. (Truncated state preparation) Let $\beta>0$, $\beta\neq 1$, and let $n_0, n_1\in\mathbb{N}$ with $n_1-n_0=2^k$ for some integer $k>0$. Define

$$|\psi_0\rangle_1=\frac{1}{C_1}\sum_{n=n_0}^{n_1-1}n^{-\beta/2}\,|n-n_0\rangle,\ C_1=\sqrt{\sum_{n=n_0}^{n_1-1}n^{-\beta}}. \tag{55}$$

Then $|\psi_0\rangle_1$ can be prepared on a standard gate-based quantum computer to precision $\varepsilon>0$, using a number of gates and ancilla qubits bounded by

$$\mathrm{Poly}(\log\varepsilon^{-1},\ \log n_1,\ \log|1-\beta|^{-1},\beta,\upsilon), \tag{56}$$

where υ denotes the number of bits of precision used to represent β. The parameters are required to satisfy

$$c_{\mathrm{ang}}=\left\lceil\frac{1}{2}\log_{2\pi}\left(\frac{192k^2n_1^{\beta}}{\varepsilon^2}\right)\right\rceil,\quad n_0>\lceil\beta+2c_{\mathrm{ang}}\rceil. \tag{57}$$

Proof sketch. The construction adapts Grover's amplitude splitting method [47], replacing integrals with partial summations. The desired distribution is generated by k iterative amplitude splits. At step m $(0\le m\le k-1)$, each bit string $w\in\{0,1\}^m$ is divided into branches w_0 and w_1, with a corresponding rotation angle $\theta_{w,m}$. All rotations for strings of the same length m can be applied in parallel; hence, the main resource cost arises from constructing the angle-preparation oracle U_m.

Theorem 2. (Angle-preparation oracle U_m) The unitary operator U_m implements

$$\sum_{w\in\{0,1\}^m}\alpha_w|w\rangle|0\rangle\mapsto\sum_{w\in\{0,1\}^m}\alpha_w|w\rangle|\theta_{w,m}\rangle, \tag{58}$$

where the rotation angle $\theta_{w,m}$ is defined by

$$\theta_{w,m}=\arcsin\sqrt{S(a_1,b_1,\beta)/S(a_2,b_1,\beta)}, \tag{59}$$

in which $S(a,b,\beta)=\sum_{n=a}^{b}n^{-\beta}$ and

$$\begin{aligned}a_1&=2^{k-m-1}(2w+1)+n_0,\\ b_1&=2^{k-m-1}(2w+2)-1+n_0,\\ a_2&=2^{k-m}w+n_0.\end{aligned} \tag{60}$$

Then U_m can be implemented such that each rotation angle is estimated to within error ε/k, using

$$\mathrm{Poly}(\log\varepsilon^{-1},\ \log n_1,\ \log|1-\beta|^{-1},\ \beta,\ \upsilon) \tag{61}$$

gates and ancilla qubits, where υ denotes the number of bits of precision used to represent β.

Proof sketch. Each angle $\theta_{w,m}$ is obtained by a bit-wise bisection procedure. At iteration i, a trial angle $\theta_{g,i}$ is represented with i bits, and $\sin^2(\theta_{g,i})$ is compared to the target ratio $S(a_1,b_1,\beta)/S(a_2,b_1,\beta)$. This comparison determines the i-th binary digit of $\theta_{w,m}$. Repeating this procedure for $\mathcal{O}(\log(k/\varepsilon))$ iterations successively refines the approximation, yielding $\theta_{w,m}$ with precision ε/k. Furthermore, we show that both the computation of $\sin^2(\theta_{g,i})$ and the partial sums can be carried out with polynomial resources, which completes the proof.

Theorem 3. (Initial state preparation) Let $\beta>0, \beta\neq 1$ and $N\in\mathbb{N}$. Define

$$|\psi_0\rangle=\frac{1}{C}\sum_{n=1}^{N}n^{-\beta/2}|n\rangle,\qquad C=\sqrt{\sum_{n=1}^{N}n^{-\beta}}. \tag{62}$$

Then $|\psi_0\rangle$ can be prepared deterministically on a quantum computer to precision $\varepsilon>0$. The required number of gates and ancilla qubits is bounded by

$$\mathrm{Poly}(\log\varepsilon^{-1},\ \log N,\ \log|1-\beta|^{-1},\beta,\upsilon), \tag{63}$$

where υ denotes the number of bits of precision used to represent β.

Proof sketch. The construction proceeds in three steps. First, prepare the auxiliary state using controlled rotations[48], noting that n_0 can be chosen to be $\mathrm{Poly}(\beta,\ \log N,\ \log(1/\varepsilon))$

$$|\psi_0\rangle_2=\frac{1}{C_2}\sum_{n=1}^{n_0-1}n^{-\beta/2}|n\rangle. \tag{64}$$

Second, construct $|\psi_0\rangle_1$ using Theorem 1, and combine it with $|\psi_0\rangle_2$ via the LCU method, yielding

$$|\psi_0\rangle_1+|\psi_0\rangle_2. \tag{65}$$

Third, eliminate the states with index $n>N$, given by

$$|\psi_0\rangle_3=\frac{1}{C_3}\sum_{n=N+1}^{n_1-1}n^{-\beta/2}|n\rangle, \tag{66}$$

yielding the desired state

$$|\psi_0\rangle=|\psi_0\rangle_1+|\psi_0\rangle_2-|\psi_0\rangle_3. \tag{67}$$

With appropriate choices of n_0 and n_1, the fixed-point amplification step detailed in Supplementary Note 5 converts the preparation into a deterministic procedure that outputs a state ε-close to the target.

Construction of the evolution operator

Theorem 4. (Evolution operator construction) Define the time evolution operator

$$U(t)=\mathrm{e}^{-i\mathcal{H}_0 t},\qquad \mathcal{H}_0=\sum_{n=1}^{N}\ln n|n\rangle\langle n|, \tag{68}$$

where the evolution time t is specified with u bits of precision. Then $U(t)$ can be implemented to precision ξ using

$$\mathrm{Poly}(\log N,\ \log|t|,\ \log\xi^{-1},u) \tag{69}$$

gates and ancilla qubits.. *Proof sketch*. The construction begins by using the logarithm oracle from Lemma 2 to compute the base-2 logarithm,

$$\sum_{n=1}^{N}\alpha_n|n\rangle|0\rangle\longmapsto\sum_{n=1}^{N}\alpha_n|n\rangle|\widetilde{\log n}\rangle, \tag{70}$$

where $\widetilde{\log n}$ approximates $\log_2 n$ to accuracy $\mathcal{O}(\xi/|t|)$. We then multiply this register by $(\ln 2)t$ to obtain an approximation to $t\ln n$. Controlled R_z rotations conditioned on the phase register then imprint the phase factor $e^{-it\ln n}$ onto each computational basis state $|n\rangle$, thereby realizing $U(t)$ to precision ξ (see Supplementary Note 6 for details).

The implementation of the controlled evolution

$$\exp(-i\mathcal{H}_{c2}t)=|\uparrow\rangle\langle\uparrow|\otimes I+|\downarrow\rangle\langle\downarrow|\otimes U(t). \tag{71}$$

in the first system, as well as the time evolution

$$\exp\left(-i\int\mathcal{H}_c dt\right)=H(\mathrm{e}^{i\theta(t)}|\uparrow\rangle\langle\uparrow|\otimes U(t)+\mathrm{e}^{-i\theta(t)}|\downarrow\rangle\langle\downarrow|\otimes U^{\dagger}(t))H \tag{72}$$

in the second system, exhibit the same complexity scaling. According to Theorem 4, both evolutions can be implemented to precision ξ using

$$\mathrm{Poly}\left(\log N,\ \log|t|,\ \log\xi^{-1},u\right) \tag{73}$$

gates and ancilla qubits.

Complexity bounds for evaluating $\zeta(s)$ in the critical strip

In classical computation, evaluating $\zeta(s)$ at large imaginary parts in the critical strip $0<\beta<1$ is notoriously challenging. The Riemann–Siegel formula can be expressed as:

$$\zeta(s)=\sum_{n=1}^{x}\frac{1}{n^s}+\chi(s)\sum_{n=1}^{y}\frac{1}{n^{1-s}}+\mathcal{O}(x^{-\beta})+\mathcal{O}(|t|^{1/2-\beta}y^{\beta-1}), \tag{74}$$

with $2\pi xy=|t|$ and $\chi(s)=2^s\pi^{s-1}\sin\left(\frac{\pi s}{2}\right)\Gamma(1-s)$. Setting $x=y=\sqrt{|t|/2\pi}$ yields

$$\zeta(s)=\sum_{n=1}^{N}n^{-s}+\chi(s)\sum_{n=1}^{N}n^{-(1-s)}+O(|t|^{-\beta/2}), \tag{75}$$

where $N=\lceil\sqrt{|t|/2\pi}\rceil$. Thus, the main computational challenge reduces to evaluating partial Dirichlet sums.

Using the quantum constructions in previous sections, we prepare the initial state

$$|\psi_0\rangle=\frac{1}{C}\sum_{n=1}^{N}n^{-\beta/2}|n\rangle,\qquad C=\sqrt{\sum_{n=1}^{N}n^{-\beta}}, \tag{76}$$

and apply the evolution operator

$$U(t)=\mathrm{e}^{-i\mathcal{H}_0 t},\qquad \mathcal{H}_0=\sum_{n=1}^{N}\ln n|n\rangle\langle n|. \tag{77}$$

The average accumulated phase factor can be obtained from the expectation values of $\sigma_x+i\sigma_y$ on an introduced probe qubit, yielding

$$\mathcal{L}(\beta,t)=\langle\psi_0|\mathrm{e}^{-i\mathcal{H}_0 t}|\psi_0\rangle=\frac{1}{C^2}\sum_{n=1}^{N}n^{-s}. \tag{78}$$

For $0<\beta<1$, we have

$$C^2=\sum_{n=1}^{N}n^{-\beta}\le 1+\int_1^N x^{-\beta}dx<\frac{N^{1-\beta}}{1-\beta}, \tag{79}$$

and $|\chi(s)|=\Theta(|t|^{1/2-\beta})$ (see Supplementary Note 7). Hence, to approximate $\zeta(\beta+it)$ with precision δ, it suffices to estimate the two partial sums $\sum_{n=1}^{N}n^{-s}$ and $\sum_{n=1}^{N}n^{-(1-s)}$ to precisions $\mathcal{O}(\delta)$ and $\mathcal{O}(\delta)|t|^{\beta-1/2}$ respectively. Therefore, the required precisions in the estimation of $\langle\sigma_x\rangle$ and $\langle\sigma_y\rangle$ of the probe qubits are

$$d_1=(1-\beta)\mathcal{O}(\delta)/N^{1-\beta},\ d_2=\beta\mathcal{O}(\delta)|t|^{\beta-\frac{1}{2}}/N^{\beta}, \tag{80}$$

respectively. To achieve an overall precision of d_1 and d_2, the preparation of the initial state, the time evolution, and sampling the outputs must each be implemented to accuracies of order $\mathcal{O}(d_1)$ and $\mathcal{O}(d_2)$, respectively. By Theorem 3 and Theorem 4, substituting ε and ξ with $\mathcal{O}(d_1)$ and $\mathcal{O}(d_2)$ yields gate and ancilla resource requirements for initial state preparation and evolution bounded by

$$\mathrm{Poly}(\log\delta^{-1},\ \log|t|,\ \log(1-\beta)^{-1},\ \log\beta^{-1},v), \tag{81}$$

and

$$\mathrm{Poly}(\log\delta^{-1},\ \log|t|,\ \log(1-\beta)^{-1},\ \log\beta^{-1},u), \tag{82}$$

respectively, where u and v denote the numbers of bits of precision used to represent t and β.

Let $\zeta'(s)$ denote the derivative of $\zeta(s)$, which can be bounded by $\mathrm{Poly}(|t|,\ |1-\beta|^{-1},\ \beta^{-1})$ (see Supplementary Note 8 for details). Thus, setting u, v as $\mathrm{Poly}(\log\delta^{-1},\ \log|t|,\ \log(1-\beta)^{-1},\ \log\beta^{-1})$ ensures that rounding errors contribute at most $\mathcal{O}(\delta)$. Substituting these bounds into the previous bounds gives gate and ancilla qubit costs $\mathcal{R}_{c1}$ and $\mathcal{R}_{c2}$ for both summations

$$\mathrm{Poly}(\log\delta^{-1},\ \log|t|,\ \log(1-\beta)^{-1},\ \log\beta^{-1}). \tag{83}$$

If $\langle\sigma_x\rangle$ and $\langle\sigma_y\rangle$ are estimated by direct sampling, sample complexity scales as $\mathcal{O}(d_1^{-2})$ and $\mathcal{O}(d_2^{-2})$. With quantum amplitude estimation, this can be reduced to $\mathcal{O}(d_1^{-1})$ and $\mathcal{O}(d_2^{-1})$. Since $N=\Theta(\sqrt{|t|})$, the required number of samples is bounded by

$$\begin{aligned}\mathcal{R}_{s1}&=\mathcal{O}(\delta^{-1})(1-\beta)^{-1}|t|^{(1-\beta)/2},\\ \mathcal{R}_{s2}&=\mathcal{O}(\delta^{-1})\beta^{-1}|t|^{(1-\beta)/2}.\end{aligned} \tag{84}$$

Hence, the overall complexity $\mathcal{R}_o=\mathcal{R}_{c1}\mathcal{R}_{s1}+\mathcal{R}_{c2}\mathcal{R}_{s2}$ is bounded by

$$\delta^{-1}|t|^{(1-\beta)/2}\mathrm{Poly}(\log\delta^{-1},\ \log|t|,(1-\beta)^{-1},\beta^{-1}). \tag{85}$$

Next, we consider the task of locating and verifying nontrivial zeros. These zeros are symmetric with respect to the critical line $\beta=\frac{1}{2}$, and there is a zero-free region, the Korobov–Vinogradov zero-free

region, which takes the form[41]

$$\beta > 1 - \frac{c}{(\ln|t|)^{2/3}(\ln\ln|t|)^{1/3}}, \tag{86}$$

in which $c > 0$ is an explicit constant. Thus, we focus on the region satisfying

$$\frac{1}{2} \le \beta \le 1 - \frac{c}{(\ln|t|)^{2/3}(\ln\ln|t|)^{1/3}}, \tag{87}$$

for which the complexity bound simplifies to

$$\delta^{-1}|t|^{(1-\beta)/2}\mathrm{Poly}(\log\delta^{-1}, \log|t|). \tag{88}$$

Data availability

The source data generated in this study have been deposited in the Zenodo repository under https://doi.org/10.5281/zenodo.20590665[49]. The source data include the numerical datasets, processed experimental data, theoretical data, and data used to generate the figures and tables. The source data are publicly available and are also available in the GitHub repository at https://github.com/lqf2025/Riemann-data.

Code availability

The code used in this study has been deposited in the Zenodo repository under DOI: 10.5281/zenodo.20590665[49]. The code used to generate and analyze the data is publicly available and is also available in the GitHub repository at https://github.com/lqf2025/Riemann-data.

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Author contributions
T.X., G.L.L. and S.W. supervised the project. S.W. proposed the initial theoretical idea and developed the theoretical framework together with Q.L. and T.X. Y.Z. performed the experiments under the supervision of T.X. W.Y., P.G. and F.N. contributed to the theoretical analysis, while C.W. and J.S. contributed to the experimental work. All authors participated in the preparation of the manuscript.

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